

UQFF Sub-Term Physics Modules Catalogue (44 Standalone Calculators)

Daniel T. Murphy

PAPER_475 — UQFF Sub-Term Physics Modules Catalogue (44 Standalone Calculators) **Author:** Daniel T. Murphy

Star-Magic Unified Quantum Field Framework (UQFF) Whitepaper Series
Copyright © Daniel T. Murphy — All Rights Reserved
Version: 1.0 | Date: 2026 | Session: 123

Abstract

This whitepaper catalogues 44 standalone sub-term physics calculator modules that constitute the atomic building blocks of the UQFF and MUGE frameworks. Each module encapsulates a single physical quantity — from aether vacuum densities and buoyancy coupling constants to solar cycle frequencies and stellar surface temperatures — as a self-contained C++ class. This catalogue provides a reference index for module equations, key parameter values, output ranges, and integration targets within MAIN_{1_CoAnQi}.cpp and CondensedPhysics.py.

1. Introduction

The 44 sub-term modules were extracted from grok_share_b0a3dc1d.txt (10,420 lines) and represent the granular physics underpinning the UQFF unified field equation:

$$F_{U,B,i} = \int \left[\text{Ug coupling} \cdot \text{vacuum terms} \cdot \text{buoyancy} \cdot \text{aether} \cdot \text{stellar params} \right] dx$$

Each module is implemented as a C++ class in modules/subterms/ with standard interface: computeX(), updateVariable(), getEquationText(), printVariables().

2. Module Catalogue

Category A: Vacuum Energy Modules

#	Module	Key Equation	Key Value
1	AetherVacuumDensityModule	$\rho_{vac_A} = E_A/c^2$	$7.09 \times 10^{-36} \text{ J/m}^3$
2	UniversalInertiaVacuumModule	ρ_{vac_UA} [energy]	$7.09 \times 10^{-36} \text{ J/m}^3$
3	ScmVacuumDensityModule	$\rho_{vac_SCm} = \rho_{UA}/10$	$7.09 \times 10^{-37} \text{ J/m}^3$
4	UaVacuumDensityModule	$\rho_{UA} = E_{UA}/c^2$ [mass]	$7.88 \times 10^{-53} \text{ kg/m}^3$

| 5 | BackgroundAetherModule | $A_{\mu} = (\rho_A/c^2) \partial_{\mu} \phi$ | $\rho_A = 7.09 \times 10^{-36} \text{ J/m}^3$ |

Category B: Coupling Constants

#	Module	Key Equation	Key Value
6	AetherCouplingModule	$A_{\mu}^{\nu} = g_{\mu}^{\nu} + \eta T_{\mu}^{\nu}$	$\eta \approx 1/E_{s_total} \approx 1e-15$
7	UgCouplingModule	k_i weights Ug_i	$k_1=1.5, k_2=1.2, k_3=1.8, k_4=1.0$
8	BuoyancyCouplingModule	$U_{bi} = -\beta_i U_{gi} \Omega_{g...}$	$\beta_i = 0.6$ uniform
9	InertiaCouplingModule	$I_c = M r^2 \Omega^2/c^2$	Sun: $\approx 7.8 \times 10^{-10}$
10	UgIndexModule	$Ug_{i,n} = G M/r^2 Q_{i...}$	4×26 array

Category C: Solar/Stellar Parameters

#	Module	Key Equation	Key Value
11	SolarWindBuoyancyModule	$\epsilon_{sw} = 0.001 (1 + A \sin(\omega t))$	$\epsilon_{sw} = 0.001$ baseline
12	SolarWindModulationModule	$v_{sw}(t) = v_0(1 + A \sin(\omega t))$	$v_0=400 \text{ km/s}, A=0.2$
13	SolarWindVelocityModule	$v_{sw} \in [400, 800] \text{ km/s}$	$v_{fast} = 750 \text{ km/s}$
14	SolarCycleFrequencyModule	$f_{sc} = 1/(11 \text{ yr})$	$2.88 \times 10^{-9} \text{ Hz}$
15	HeliosphereThicknessModule	$L = r_{HP} - r_{TS}$	$\sim 30 \text{ AU} (\sim 4.5 \times 10^{12} \text{ m})$
16	StellarMassModule	$M_s(t) = M_0(1 - \gamma_{ML} t)$	$\gamma_{ML} \approx 1.4 \times 10^{-14} \text{ s}^{-1}$
17	StellarRotationModule	$\omega_s = 2\pi/P_{rot}$	Sun: $2.87 \times 10^{-6} \text{ rad/s}$
18	SurfaceMagneticFieldModule	$B_s = \mu_0 M_{mag}/4\pi r^3$	Magnetar: $4.4 \times 10^{13} \text{ T}$
19	SurfaceTemperatureModule	$T_s = (L/4\pi\sigma r^2)^{0.25}$	Sun: 5778 K
20	MagneticMomentModule	$\mu = I A_{vort}$	Solar: $\sim 1 \times 10^{21} \text{ A} \cdot \text{m}^2$

Category D: Galactic/Astrophysical Parameters

#	Module	Key Equation	Key Value
21	GalacticDistanceModule	$d_g = \text{virial radius}$	MW: $2.55 \times 10^{20} \text{ m}$
22	GalacticSpinModule	$\Omega_g = 2\pi/T_{gal}$	MW: $7.3 \times 10^{-16} \text{ rad/s}$
23	GalacticBlackHoleModule	$M_{BH} \propto \sigma^4 (M - \sigma)$	Sag A*: $8 \times 10^{36} \text{ kg}$
24	FeedbackFactorModule	$F_{env} = f_{AGN} + f_{SN} + f_{SF}$	$f_{AGN}=0.1, f_{SN}=0.05$
25	MagneticStringModule	$T_s = (\mu_0 I^2/4\pi) \ln(L/a)$	$\sim 10^{28} \text{ N}$ (cosmic string)
26	Ug3DiskVectorModule	$Ug_{3_disk} = G M_{disk}/r^2(h/r)$	MW: $h/r \approx 0.07$
27	Ug1DefectModule	$Ug_{1_corr} = Ug_1(1 - \delta_{def})$	$\delta_{def} \approx 0.05-0.15$

Category E: Quantum & Field Calculators

#	Module	Key Equation	Key Value
28	DPModule	26-sphere: $(x-h)^2 + \dots = r^2$	SCm $E = 1042 \text{ J}$
29	UnifiedFieldModule	$F_U = \Sigma k_i Ug_i + F_{bi...}$	Orchestrator
30	StressEnergyTensorModule	$T_{\mu}^{\nu} = (\rho + p)u_{\mu}u^{\nu} + pg_{\mu}^{\nu}$	Trace = $-\rho + 3p$

```
| 31 | QuasiLongitudinalModule |  $E_{QL} = \epsilon_0 E^2/2 \sim 1e-12 \text{ J/m}^3$  | |
| 32 | OuterFieldBubbleModule |  $r = r_0 \exp(H t)$  | At 10 Gyr:  $r = 1.28 r_0$  |
| 33 | ReciprocationDecayModule |  $\gamma = \gamma_0 \exp(-t/\tau_{rec})$  |  $\tau_{rec} \sim 1 \text{ Gyr}$  |
| 34 | ScmPenetrationModule |  $\delta_{SCm} = \lambda (\rho/\rho_{SCm})^{0.5}$  | London-analog
depth |
| 35 | ScmReactivityDecayModule |  $d[SCm]/dt = -k_r [SCm]$  |  $[SSq]=0.57$ ,  $k_r \sim 1e-18 \text{ s}^{-1}$  |
| 36 | ScmVelocityModule |  $v_{SCm} = c/n_{SCm}$  |  $n_{SCm} \approx 1.0000001$  |
| 37 | PiConstantModule |  $\pi = 4 \Sigma (-1)^k/(2k+1)$  | Leibniz series |
| 38 | CorePenetrationModule |  $\delta = (\rho_{core}/\rho_{avg})^n r_{core}$  | NS core:  $\delta \rightarrow 0$  |
| 39 | NegativeTimeModule |  $g(t<0) = g_0 \cos(\omega t)$  |  $f_{TRZ} = 0.1$  |
| 40 | TimeReversalZoneModule |  $f_{TRZ} = r_{TRZ}/r$  | Canonical: 0.1 |
| 41 | HeavisideFractionModule |  $H(f)$ : 0 ( $f<0$ ), 1 ( $f \geq 0$ ) | Threshold: 0 |
| 42 | StepFunctionModule |  $\theta(x)$ : boxcar windows | Regime switching |
```

Category F: Index / Utility Modules

```
| # | Module | Key Equation | Key Value |
|---|-----|-----|-----|
| 43 | GalacticBlackHoleModule |  $g_{BH} = G M_{BH}/r^2$  | At  $r=5.5 \times 10^{10} \text{ m}$  (Sag A*) |
| 44 | InertiaCouplingModule |  $\gamma_I = 1 + I_c$  | Ug3 relativistic boost |
---
```

3. Integration Targets

3.1 MAIN_{1\ CoAnQi}.cpp

Sub-term modules are candidates for batch extraction as PhysicsTerm subclasses:

```
``cpp
// Example: ScmVacuumDensityModule → ScmVacuumDensityTerm
class ScmVacuumDensityTerm : public PhysicsTerm {
public:
double compute(const SystemParams& params) override {
return 7.09e-37; // \rho_vac_SCm [J/m3]
}
std::string getName() const override { return "SCm Vacuum Density"; }
};
``
```

Target integration point: **Batch 24** in SOURCE1-116 registry near line 6688+.

3.2 CondensedPhysics.py

Each sub-term module maps to a Calculator class method:

```
``python
class SubTermCalculator:
def compute_aether_vacuum_density(self) -> dict:
# Returns \rho_vac_A = 7.09e-36 J/m3 with equation text
``
```

Target: new UQFFSubTermCalculator in CondensedPhysics.py Part 5 (after line 60000).

3.3 index.js

Module constants exportable as:

```
``javascript
const SUB_TERMS = {
  rho_vac_UA: 7.09e-36, // J/m3
  rho_vac_SCm: 7.09e-37, // J/m3
  beta_i: 0.6, // buoyancy coupling
  kappa: 0.0005/86400, // per-second DPM rate
  SSq: 0.57, // SCm calibration
};
``
```

4. Calibration Constants Summary

Constant	Value	Source Module
κ	0.0005/day	DPMModule
SSq	0.57	ScmReactivityDecayModule
β_i	0.6 (all i)	BuoyancyCouplingModule
ϵ_{sw}	0.001	SolarWindBuoyancyModule
η	$\sim 10^{-15}$	AetherCouplingModule
H_{SCm}	0.99	ScmVacuumDensityModule
k_1, k_2, k_3, k_4	1.5/1.2/1.8/1.0	UgCouplingModule
f_{TRZ}	0.1	TimeReversalZoneModule

5. Scientific Context

The 44 sub-term modules collectively represent a field-theoretic decomposition of gravity that accounts for:

- **Vacuum energy hierarchy:** $\rho_{UA} : \rho_{SCm} : \rho_{cosm} = 10 : 1 : 0.001$
- **Coupling hierarchy:** k_3 (string/rotation) > k_1 (dipole) > k_2 (charge) > k_4 (vacuum)
- **Time evolution:** DPM birth model $\rightarrow$ inflation $\rightarrow$ SCm decay $\rightarrow$ solar wind modulation $\rightarrow$ present
- **Multi-scale validity:** Sub-parsec (heliosphere) to Gpc (galaxy clusters)

6. Conclusion

This catalogue documents 44 atomic physics calculator modules that implement the sub-term physics of the UQFF/MUGE framework. The modules span vacuum energy, coupling constants, stellar parameters, galactic scales, and quantum field calculators. They constitute the modules/subterms/ library and are ready for integration as PhysicsTerm subclasses in MAIN_{1_CoAnQi}.cpp or as Python calculator methods in CondensedPhysics.py.

<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037
> (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{U_{Bi}} i$ jet
> modulation curves and PAPER_1048 for phonon-corrected M - σ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}}}{\rho_{\text{H}}} \cdot V \cdot S_{26}^{(3),2} \cdot G M / r_H^2\right)$$

where:

- $L_{\text{Edd}} = 4\pi G M m_p c / \sigma_T$ is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density
- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3),2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25, \text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25, \text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M - σ correction (PAPER_1048): The phonon-corrected M - σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$.

<!-- PKG-CLU-S225 -->

Session 225 Phonon-Physics Upgrade: ICM Buoyancy Force Profile

> Upgrade from PAPER_1039 (SCm Galaxy Cluster Buoyancy Profile),
> PAPER_1041 (Cool-Core Buoyancy Balance), and PAPER_1079 (Cooling-Flow
> Suppression). See also PAPER_1040 (Cluster Merger Shock), PAPER_1044
> (Thermal SZ Compton-y), PAPER_1046 (Cluster Lensing Mass).

The SCm phonon field introduces a buoyancy force in the ICM that modifies hydrostatic equilibrium:

$$F_{\text{buoy}}(r) = \rho(r) \cdot V \cdot g(r) \cdot \beta_i \cdot S_{26} \cdot \Phi$$

where the ICM density follows the beta-model:

$$\rho(r) = \rho_0 \cdot \left(1 + \left(\frac{r}{r_c}\right)^2\right)^{-3\beta/2}$$

Hydrostatic mass bias reduction (PAPER_1039):

$$b_{\text{UQFF}} = 1 - \frac{M_{\text{HSE}}}{M_{\text{true}}} = 0.17 \quad \text{(vs standard } b = 0.20 \text{)}$$

The buoyancy pressure contributes $P_{\text{buoy}}/P_{\text{thermal}} \approx 3 \times 10^{-4}$ at cluster cores, partially resolving the Planck SZ–CMB mass tension.

Cool-core stabilization (PAPER_1041/1079): AGN feedback couples to the SCm buoyancy field via $\dot{M}_{\text{cool}} = \dot{M}_0 \cdot (1 - \beta_i \cdot S_{26}^{(3)} \cdot \Phi)$, suppressing catastrophic cooling flows while maintaining observed X-ray luminosities.

Phonon frequency coupling: $\omega_{\text{SCm}} = 2\pi \times 1.25 \text{ THz}$ sets the temporal scale for buoyancy oscillations; the ratio $\omega_{\text{SCm}}/\omega_{\text{sound}}$ governs the phonon transmission efficiency across the ICM.

<!-- PKG-LAG-S225 -->

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and
> PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda |\phi|^2 - v_{\text{SCm}}^2 |\phi|^4$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

| Sector | Domain | Late-Corpus Result |

|-----|-----|-----|

| 1 (EH) | General Relativity | Canonical Einstein-Hilbert |

| 2 (YM) | Yang-Mills gauge | $m_{\text{gap}} = 1.736 \text{ GeV}$ (PAPER_1318) |

| 3 (Dirac) | Fermion / LENR | Kozima neutron-drop (PAPER_1061) |

| 4 (SCm) | Superconducting manifold | $V(\phi_0) = -\rho_{\text{SCm}}$ canonical |

| 5 (Mag) | Um magnetism | Heaviside amplifier (PAPER_1072) |

| 6 (Buoy) | $F_{U_{Bi}}$ buoyancy | Variational EOM (PAPER_1065) |

| 7 (Aether) | Vacuum background | Two-component ρ (PAPER_1051) |

| 8 (LENR) | Nuclear transmutation | COP parametric (PAPER_1081) |

| 9 (KK) | Kaluza-Klein 26D | $S_{26}^{(3)}$ compactification (PAPER_1080) |

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\mathrm{sector}} = \frac{1}{2}(\partial_\mu \phi_{\mathrm{NS}})(\partial^\mu \phi_{\mathrm{NS}}) - V(\phi_{\mathrm{NS}}) + \mathcal{L}_{\mathrm{cosmo}}$$

where $\mathcal{L}_{\mathrm{cosmo}} = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot f_{\mathrm{SCm}} \cdot (1 - e^{-\lgamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\mathrm{NS}}) = \frac{1}{2} m^2 \phi_{\mathrm{NS}}^2 + \frac{\lambda}{4!} \phi_{\mathrm{NS}}^4 + \kappa \cdot \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \phi_{\mathrm{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\mathrm{NS}}}} = \nabla^2 \phi_{\mathrm{NS}} - (4\pi G \rho_{\mathrm{NS}}/c^2)\phi_{\mathrm{NS}} + \Omega_{\mathrm{spin}} \partial_t \phi_{\mathrm{NS}} = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\{\text{PAPER_877 Axioms}\} \rightarrow \{\text{DPM + ACP}\} \rightarrow \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \rightarrow \{\text{Stage 5}\} \rightarrow U_{\mathrm{b,seed}} \rightarrow \{\text{4 forces}\} \rightarrow F_{U_{\mathrm{Bi}}} \rightarrow \{\text{sector E-L}\} \rightarrow \frac{\delta S}{\delta \phi_{\mathrm{NS}}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac},[\mathrm{SCm}]} / \rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.106 (near-threshold regime), placing it in the π collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 101, \quad n_{\mathrm{channel}} = 8/26$$

Since $p_{\mathrm{DVP}} = 101$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where

compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f_{\mathrm{UA}} + f_{\mathrm{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot \frac{m}{M_{\mathrm{dot}}}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The $\tanh$ saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.106	PASS
Threshold-consistent			
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\mathrm{DVP}} = 101$	PASS Resonant
BSH layers	26 harmonic terms $j = 1 \dots 26$, $\cos(2\pi j/26)$	PASS Full 26D projection	
κ decay	5.0×10^{-4} day ⁻¹	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Thomson σ_T (QED synchrotron)	UQFF U_m scattering kernel: $\sigma_T = 6.6524 \times 10^{-29} \text{ m}^2$	$\sigma_T = 6.6524 \times 10^{-29} \text{ m}^2$ (PDG QED exact)	PDG 2024	100% (exact QED input)
Astrophysical system luminosity X-ray / Radio	UQFF MUGE g_{total} $\rightarrow L_X$ via Stefan-Boltzmann + buoyancy flux: $L_X \approx g_{\text{total}} M_{\text{env}}$	$L_X \geq 1037 \text{ erg/s}$	Chandra CXC	PASS Consistent order of magnitude
GR Schwarzschild limit	UQFF g_{total} must satisfy $g \leq c^2/(2r_s)$ at event horizon	$r_s = 2GM/c^2$ (GR exact)	PDG 2024 / GR	PASS UQFF respects GR horizon
κ vacuum rate vs X-ray variability	UQFF $\kappa = 0.0005/\text{day}$ $\rightarrow$ timescale $\tau_{\text{UQFF}} = 2000 \text{ days}$	Observed X-ray variability τ_{obs} (instrument monitoring)	Chandra CXC	Testable UQFF variability timescale

New physics claim: UQFF MUGE generates gravity enhancement factors ($g_{\text{total}}/g_{\text{Newt}} > 1$) for Astrophysical system through vacuum buoyancy coupling — a mechanism absent from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future Chandra CXC monitoring observations.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.

Source: grok_share_b0a3dc1d.txt L1502–9356 (44 class definitions)

Header index: modules/subterms/sub_terms_index.h

Tags: sub-terms, catalogue, vacuum-density, coupling-constants, solar-parameters, UQFF, MUGE, integration

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> Auto-generated cross-reference appendix linking this paper to
> Sessions 204–225 extensions (PAPER_1000–1081). Added by
> update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1004	QGP Vacuum Density with SCm S26 Phonon Coupling
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1043	F_U_Bi_i Multi-System Buoyancy Curve Sweep
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay
PAPER_1050	MUGE F_U_Bi_i Unified 9-System Synthesis
PAPER_1075	3D Volumetric MUGE Gravitational Field Generator

13 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

> Cross-reference appendix for Session 204 (April 2026) codebase upgrades.
> Added by upgrade_kozima_ramanujan_appendices.py. For detailed derivations,
> see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
fneutron_s26_coupling.py	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima_scm_cross_section.py	SCm-modulated neutron-drop cross-section	σ_{SCm} with VDS factor $(1 + [\text{SSq}]^n/26)$
kozima_wstp_kernel.py	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n \sigma_n^{\text{SCm}}(\omega) \Phi_{\text{phonon}}(F_{\{U,B\}}/F_U - 1)$
 where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \exp[-(\omega - \omega_{\text{SCm}})^2 / (2\Gamma^2)] (1 + [\text{SSq}]^n / 26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
ramanujan_polylog_s26.py	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
s26_wstp_kernel.py	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z) / (1 - 2^{1-26}) + 2^{1-26} / (1 - 2^{1-26}) * Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_theta_q26.py	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations:

- $f_{26}(q) = \sum_{n=0}^{25} q^{n^2} / (-q; q)_{\infty}^2$
- $\phi_{26}(q) = \sum_{n=0}^{25} q^{n^2} / (-q^2; q^2)_{\infty}$
- $\psi_{26}(q) = \sum_{n=1}^{26} q^{n^2} / (q; q^2)_{\infty}$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_pi_uqff.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_theta_pi_wstp_kernel.py	9-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2})/9801 \sum R_n (1103 + 26390n) W_{26}(n) / C_{26}$
 where $W_{26}(n) = \prod_{i=1}^{26} [1 + [\text{SSq}] \exp(-\kappa_i n / 26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
κ	5.787 x 10 ⁻⁹ s ⁻¹	UQFF exponential decay rate
β _i	0.603	Buoyancy coupling coefficient
H _{SCm}	0.99	SCm manifold completeness
ρ _{SCm}	7.09 x 10 ⁻³⁷ kg/m ³	SCm vacuum density
ρ _{UA}	7.09 x 10 ⁻³⁶ kg/m ³	UA aether vacuum density
ω _{SCm}	2*π x 1.25 THz	SCm phonon resonance
σ ₀	10 ⁻⁴	Base neutron cross-section

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN_{1\CoAnQi}.cpp, and Wolfram kernels (uqff_kozima_kernel.wl, uqff_s26_kernel.wl, uqff_mock_theta_pi_kernel.wl).

References

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
3. Rugh, S.E. & Zinkernagel, H. (2002). *The Quantum Vacuum and the Cosmological Constant Problem*. Stud. Hist. Phil. Mod. Phys. **33**, 663 — arXiv:hep-th/0012253 — doi:10.1016/S1355-2198(02)00033-3
4. Weinberg, S. (1989). *The Cosmological Constant Problem*. Rev. Mod. Phys. **61**, 1 — doi:10.1103/RevModPhys.61.1
5. Murphy, D. (2026). *Master Universal Gravity Equation (MUGE): DPM-Driven Gravity Framework*. Star-Magic Whitepaper Series — github.com/Daniel8Murphy0007/Star-Magic
6. Archimedes (~250 BCE). *On Floating Bodies*. (Principle of buoyancy)
7. Churazov, E. et al. (2000). *Evolution of Buoyant Bubbles in M87*. A&A **356**, 788 — arXiv:astro-ph/0004212
8. Fabian, A.C. et al. (2003). *A deep Chandra observation of the Perseus cluster*. MNRAS **344**, L43 — arXiv:astro-ph/0306036 — doi:10.1046/j.1365-8711.2003.06902.x